

Instruction Backup

1 Core Idea

Definition

Instruction backup is the problem of restoring the CPU state so the instruction that caused a page fault can be restarted correctly.

When a page fault occurs, the faulting instruction may have executed only partly. After the OS loads the missing page, the same instruction must be executed again from its original starting state.

Main Difficulty

Restarting an instruction is hard if the CPU cannot clearly tell the OS where the instruction began and which side effects have already happened.

2 Why Restart Is Needed

A page fault stops an instruction before it finishes. The OS handles the fault, loads the missing page, and then the process must continue.

Restart Requirement

after page fault handling, rerun the faulting instruction correctly

Correct restart means:

- the program counter points to the beginning of the instruction;
- registers have the correct values;
- partial side effects are undone if necessary.

3 Example Instruction

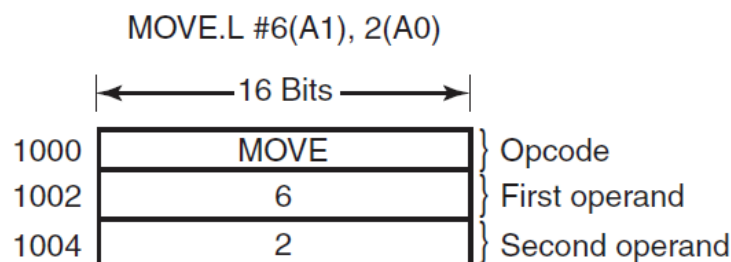


Figure 3-27. An instruction causing a page fault.

The example instruction is:

```
MOVE.L #6(A1), 2(A0)
```

It occupies 6 bytes:

Address	Contents	Meaning
1000	MOVE	Opcode.
1002	6	First operand offset.
1004	2	Second operand offset.

Figure Meaning

The instruction starts at address 1000, but a fault may occur while fetching or using different words of the instruction.

4 Program Counter Ambiguity

The instruction may make several memory references:

- fetch the opcode at 1000;
- fetch or use the first operand information at 1002;
- fetch or use the second operand information at 1004.

Depending on which reference faults, the program counter at trap time might be:

1000, 1002, or 1004

Ambiguity

If the program counter is 1002, the OS may not know whether address 1002 is the middle of an instruction that began at 1000, or the start of a different instruction.

Without extra hardware information, the OS may be unable to unambiguously find the first byte of the faulting instruction.

5 Why Instruction Decoding Can Be Hard

Some CPUs have variable-length instructions. An instruction may contain:

- opcode bytes;
- operand specifiers;
- address offsets;
- immediate constants.

So a word in memory can mean different things depending on where instruction decoding began.

Key Point

The OS cannot safely restart an instruction unless it knows the exact original instruction boundary.

6 Autoincrement and Autodecrement Problems

Some addressing modes modify registers as a side effect.

Autoincrement

Autoincrement means a register is automatically increased as part of executing an instruction.

If a page fault occurs during such an instruction, the OS must know whether the increment already happened.

Case	OS requirement
Register was incremented before the fault	OS may need to decrement it before restarting.
Register was not incremented before the fault	OS must not undo anything.

Autodecrement creates the same kind of problem in the opposite direction.

CPU-Dependent Detail

Whether increment/decrement happens before or after the memory reference can depend on the instruction and CPU model.

7 Hardware Support for Restart

Some CPU designers solve this problem by saving extra internal state.

Hardware information	Why it helps
Hidden saved instruction-start PC	Tells the OS where the faulting instruction really began.
Autoincrement/autodecrement record	Tells which registers were already changed and by how much.
Instruction state registers	Help restore the CPU to a restartable state.

Good Design

If the hardware records enough restart information, the OS can undo partial effects and restart the instruction cleanly.

8 If Hardware Does Not Help

Without hardware support, the OS must reconstruct what happened.

This may require:

- reading the saved program counter;
- decoding nearby instruction bytes;
- determining which operand faulted;
- checking whether registers were already modified;
- undoing side effects manually.

Implementation Burden

If the CPU does not provide enough fault-state information, page fault recovery becomes much more complex for the operating system.

9 Final Summary

One-Box Summary

A page fault can stop an instruction halfway through.
The OS must restart the instruction from its beginning.
Variable-length instructions make the starting address ambiguous.
Autoincrement/autodecrement can leave partial register changes.
Good hardware saves enough state for clean instruction restart.